1. Standard Solution
   - (1) Totaling
   - (2) Avg
   - 3. min max
   - 4. linear search
   - 5. Bubble sort

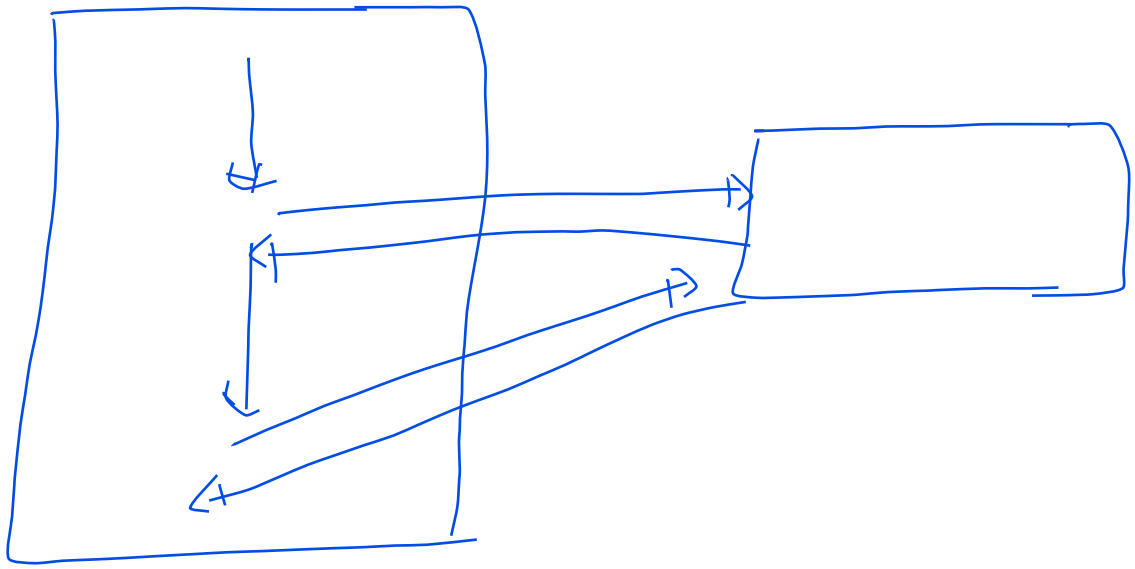
PROCEDURE AND FUNCTION

↳ ← subroutine

↳ PROCEDURE (show)

↳ FUNCTION (RETURN)

Main program (under) x



PROCEDURE EXAMPLE

```
DECLARE num1, num2, sum : INTEGER   #Global variable
INPUT num1, num2
sum ← num1 + num2
OUTPUT "The sum is", sum
```

main

```
PROCEDURE Show( )
    OUTPUT "you are the best"
ENDPROCEDURE

CALL Show( )
CALL Show( )
```

OUTPUT

you are the best

you are the best

```
PROCEDURE number( )
    DECLARE num : INTEGER
    num ← RANDOM( ) * 10
    OUTPUT "The Random value is" num
ENDPROCEDURE

CALL number( )
CALL number( )
OUTPUT num // invalid
```

LOCAL variable

PF

scope

Main

OUTPUT

Local ← Born in scope and die inside the scope

Data

Param

PROCEDURE (parameter)

→ PROCESS → SHOW

```
PROCEDURE Sum(num1 : INTEGER, num2 : INTEGER)
    DECLARE Add : INTEGER //LOCAL
    Add ← num1 + num2
    OUTPUT Add
ENDPROCEDURE

CALL Sum(5,2)
CALL Sum(10,2)
```

OUTPUT

7

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